MATRICES

TRANSPOSITION

TRANSPOSITION

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\mathbf{A}^{\mathrm{T}} = \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}$$

TRANSPOSITION

$$A = \begin{bmatrix} 2 & 4 & 6 \\ 0 & 3 & 1 \\ 1 & 9 & 8 \\ 0 & 3 & 1 \end{bmatrix} \quad A^T = \begin{bmatrix} \\ \\ \\ \end{bmatrix}$$

TRANSPOSITION

$$A = \begin{bmatrix} 2 & 6 & 1 \\ 6 & 3 & 4 \\ 1 & 4 & 8 \end{bmatrix} \qquad A^T = \begin{bmatrix} 2 & 6 & 1 \\ 6 & 3 & 4 \\ 1 & 4 & 8 \end{bmatrix}$$

✓ Symmetrical

TRANSPOSITION

$$A = \begin{bmatrix} 2 & 5 & 9 \\ 4 & 2 & 3 \\ 1 & 6 & 8 \end{bmatrix} \qquad A^T = \begin{bmatrix} 2 & 4 & 1 \\ 5 & 2 & 6 \\ 9 & 3 & 8 \end{bmatrix}$$

× Not Symmetrical

TRANSPOSITION

$$\mathbf{A} = \begin{bmatrix} 0 & 2 & -4 \\ -2 & 0 & 6 \\ 4 & -6 & 0 \end{bmatrix} \qquad \mathbf{A}^{\mathrm{T}} = \begin{bmatrix} 0 & -2 & 4 \\ 2 & 0 & -6 \\ -4 & 6 & 0 \end{bmatrix}$$

Anti-symmetrical